INCH-POUND

MIL-C-368J
4 May 1989
SUPERSEDING
MIL-C-368H
15 October 1973

MILITARY SPECIFICATION

CLOTH, SATIN, RAYON AND CLOTH, TWILL, RAYON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rayon lining cloths for use in various clothing items.

1.2 Classification. The cloth shall be of the following classes as specified (see 6.2).

Class 1 - Twill, 3.7 ounce
Class 2 - Twill, 4.2 ounce
Class 3 - Satin, 4.5 ounce

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available from the Federal Trade Commission, Washington, DC 20580-0001.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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* AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Standard sample. The dyed and finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

* 3.3 Material. The yarns shall be multifilament rayon. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.4 Color. The color of the cloth shall be as specified (see 6.2) and shall match the standard sample. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed and finished cloth shall show no more labile sulfur than shown by the standard sample when tested as specified in 4.4.3. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.4.3. When dyestuffs are used which require an additional treatment to fix the dyestuffs to the fibers, the contractor shall certify in writing to the contracting officer that in the additive material formaldehyde is not the principle fixing agent and that the additive material used will not create an odor or cause degradation of the cloth during storage.

* 3.4.1 Matching. The color of the dyed cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 $\pm$ 200 K, with illumination of 100 $\pm$ 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 $\pm$ 200 K.

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* 3.4.2 Colorfastness. The finished cloth shall show fastness to perspiration and wet drycleaning equal to or better than the standard sample or equal to or better than a rating of "good". The finished cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transferance Scale rating of not lower than 3.5 except for black, which shall have an AATCC Chromatic Transference Scale rating of not lower than 2.5. Testing shall be as specified in 4.4.3.

3.5 Physical requirements. The physical requirements of the finished cloth shall be as specified in table I when tested as specified in 4.4.3.

TABLE I. Physical requirements

Class	Weight oz/sq yd (minimum)	Yarns per inch (minimum)		Breaking strength (dry), pounds (minimum)		Breaking strength (wet), pounds (minimum)		Weave
		Warp	Filling	Warp	Filling	Warp	Filling	
1	3.7	121	67	100	50	40	20	2/1 right twill
2	4.2	142	71	115	55	46	22	2/1 right twill
3	4.5	180	67	150	55	60	22	5-harness satin (see fig. 1)

3.5.1 Width. Unless otherwise specified (see 6.2), the width shall be not less than 41 inches inclusive of selvages.

3.6 Finish. The cloth shall be natural finish. The use of resins, oils, starches, or gums in the finishing of the cloth is prohibited.

3.7 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor greater than 8.5 when tested as specified in 4.4.3.

* 3.8 Shrinkage. The shrinkage or elongation in both the warp or filling direction of the finished cloth shall not be greater than 3.5 percent for the individual sample unit and not greater than 3.0 percent for the lot average when tested as specified in 4.4.3.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards. Each length shall be put-up in a roll as specified in 5.1.

3.10 Fiber identification. Each roll shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

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3.11 Face identification. The face side of the cloth shall be identified by stamping the face side with the word "FACE" at each end of the roll.

3.12 Workmanship. The finished cloth shall conform to the quality of product established by this specification. The demerit points per 100 square yards shall not exceed the specified point values.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

* 4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2.1 through 4.4.2.4 and tested for the characteristics specified in table II.

4.4 Quality conformance inspection.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item visual examination.

4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in section III of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.4.2.1.1. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 25.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 38.0 points. If no individual roll exceeds 38.0 points per 100 square yards, the lot shall be acceptable with respect to this characteristic. If one roll exceeds 38.0 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 38.0 points per 100 square yards. For individual roll quality examination, local type defects only shall be scored. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

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The following defects, when present, shall be scored four points for each yard in which they occur:

Objectionable odor
Width less than specified
Poor dye penetration, mottled, streaky or cloudy
Bias or bowed filling - distortion at any point, 2 or more inches from normal alignment
Turned selvage, which cannot be flattened by manual pressure

4.4.2.2 Length examination. During the yard-by-yard examination, each roll in the sample unit shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a length defect. The lot shall be rejected if two or more rolls in the sample are defective with respect to length or if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets.

4.4.2.3 Shade examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade or shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present:

Face identification missing from either or both ends.

Not labeled or ticketed in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

4.4.3 End item testing. The cloth shall be tested for the characteristics listed in table II. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table II shall be followed. The sample unit shall be 3-½ continuous yards full width of the finished cloth. The lot shall be rejected if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size (yards)	Sample size (sample units)
800 or less	2
800 up to and including 22,000	3
22,001 and over	5

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TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Rayon identification	3.3	1600 1/
Yarn (multifilament)	3.3	Visual 2/
Labile sulfur	3.4	2020
Fixing agent	3.4	1/
Colorfastness to:		
Perspiration	3.4.2	5680
Wet drycleaning	3.4.2	5622
Crocking	3.4.2	5651
Weight	3.5	5041
Yarns per inch	3.5	5050
Breaking strength:		
Dry	3.5	5100
Wet	3.5	5100 3/
Weave	3.5	Visual 2/
Resins, oils, starches, gums	3.6	1/
pH	3.7	2811
Shrinkage	3.8	5580

1/ Unless otherwise specified, a certificate of compliance is required and will be acceptable for the requirements stated.

2/ One determination per sample unit, and the result reported as "pass" or "fail".

3/ Three specimens shall be immersed in tap water at room temperature for 2 hours. The specimens shall then be removed from the water one at a time and tested immediately for tensile strength. The results shall be reported as the average of the three determinations to the nearest 1 pound.

* 4.4.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1133.

* 5. PACKAGING

* 5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth shall be put-up on rolls and preserved in accordance with the applicable requirements of PPP-P-1133.

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract, purchase order, or shipments shall be marked in accordance with the applicable requirements of PPP-P-1133.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The lining cloth is intended for use in the following garments:

Class 1 - General utility use, garrison cap, service cap and hats, male and female and enlisted personnel uniforms.

Class 2 - Enlisted men's coats, male and female officers' and enlisted personnel type uniforms, raincoats with liners.

Class 3 - Officer type overcoats, dress uniforms, (men and women).

* 6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Class required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Color of cloth required (see 3.4).
f. Width of cloth required, if other than specified (see 3.5.1).
g. Minimum acceptable length if other than specified (see 3.9).
h. Selection of applicable levels of put-up, preservation and packing (see 5.1 and 5.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids.

6.4.1 Standard sample dye combinations. The standard samples have been dyed with the following dyestuff formulations:

Army Green 45

Direct Yellow 105
Direct Yellow 106
Direct Blue 191
Direct Fast Red RL

Gold 68 and Old Gold 175

Direct Blue 191
Direct Yellow 105
Direct Yellow 106

Army Tan 302

Direct Blue 191
Direct Yellow 105
Direct Fast Grey LVL

* 6.5 Subject term (key word) listing.

Fabric
Lining
Uniform

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the previous issue.

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Custodians:

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Air Force - 11

Review activities:

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Air Force - 82, 99
DLA - CT

User activity:

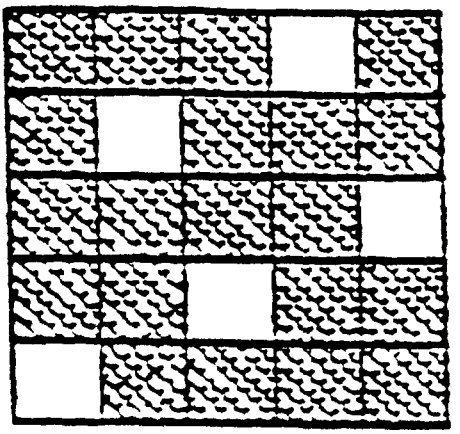
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Preparing activity:

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Class 3

FIGURE 1. Weave construction.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-368J	Cloth, Satin, Rayon and Cloth, Twill, Rayon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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